

Task 1:

Laptop No-Power Diagnostic Steps

Objective: Outline 5 systematic troubleshooting steps to diagnose a laptop that refuses to power on even when connected to an AC adapter.

1. Verify AC Power Outlet, Adapter & Cable Integrity:

- **What I am checking for:** Verifying that the wall outlet provides active AC power and that the power adapter output LED lights up. Inspecting the power cord for fraying, bends, or physical damage, and measuring DC voltage output with a multimeter to ensure it matches the adapter rating (e.g., 19V).

2. Inspect the Laptop DC Charging Jack & Port:

- **What I am checking for:** Checking for a loose, wobbly, or physically damaged DC jack/charging port. Ensuring the pin inside the port is unbroken and making solid electrical contact with the charger tip.

3. Perform a Hard Power Reset (Static Discharge):

- **What I am checking for:** Checking if residual flea power or static charge trapped in the motherboard capacitors is preventing the power management IC from initiating startup.
(Procedure: Unplug AC power, disconnect/remove the battery, press and hold the power button for 20–30 seconds, then reconnect AC power only).

4. Isolate Peripheral & RAM Components:

- **What I am checking for:** Checking if a shorted RAM module, corrupted CMOS battery (RTC), or shorted USB peripheral is blocking the power-on sequence. Unplugging all external devices and reseating the RAM sticks one by one.

5. Test Motherboard Primary Power Rail & Charging IC:

- **What I am checking for:** Using a multimeter in DC voltage and continuity modes to inspect the 19V primary rail, MOSFETs, and power management IC on the motherboard for short circuits or blown components.

Task 2:

Helpdesk Ticket: Blank Screen / No Display Troubleshooting Flow

Objective: Outline a systematic step-by-step troubleshooting flow for a helpdesk ticket where the PC powers on (fans running, lights active) but the monitor displays a blank screen.

Troubleshooting Step-by-Step Flow:

1. Verify Monitor Power and Display Cable Connections:

- Ensure the monitor's power cable is plugged in and the LED indicator is ON.
- Unplug and firmly reconnect both ends of the display cable (HDMI / DisplayPort) to ensure a secure connection.

2. **Verify Correct Monitor Input Source (OSD Settings):**

- Use the monitor's physical buttons to open the On-Screen Display (OSD) menu.
- Ensure the input source is manually set to match the active cable (e.g., set to **HDMI 1** or **DisplayPort** instead of Auto-Detect).

3. **Check Correct Display Output Port (GPU vs Motherboard):**

- If the system has a dedicated Graphics Card (GPU), ensure the HDMI/DisplayPort cable is connected directly to the **GPU output port**, not the motherboard's onboard video port.

4. **Perform Memory (RAM) Reseating:**

- Power down the PC, disconnect power cables, and open the chassis.
- Remove the RAM modules, clean the contact pins, and firmly reinsert a single RAM stick to

clear any memory POST initialization block causing the blank screen.

5. Cable & Monitor Isolation Test:

- Replace the HDMI/DisplayPort cable with a known-working cable.
- Connect the PC to an alternate monitor or TV to determine if the original monitor or its backlight has failed.

Task 3:

Wireless Mouse Fault Isolation Procedure

Objective: Systematically isolate and identify whether an unresponsive wireless mouse issue is caused by the mouse hardware itself, a faulty USB port, or operating system settings.

1. Isolating Mouse Hardware (Battery & Power Check):

- **Power & Battery Inspection:** Verify that the mouse's physical ON/OFF switch is set to ON. Replace the AA/AAA battery with a brand-new battery or connect the charging cable if it uses an integrated rechargeable battery.
 - **Optical Sensor & Dongle Pair Check:** Check if the optical sensor LED underneath lights up. Press the reset/connect button (if available) to re-sync the mouse with its 2.4GHz USB receiver.
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2. Isolating the USB Port:

- **Port Swapping:** Unplug the wireless USB dongle/receiver from its current port and insert it into a known-working USB port (preferably a direct rear port on a desktop motherboard, bypassing external USB hubs).
 - **Port Verification:** Plug the working USB keyboard or a USB flash drive into the original USB port to verify if that specific port passes power and data. If the keyboard works in that port, the USB port itself is functional.
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3. Isolating System Settings & Device Drivers:

- **Device Manager Inspection:** Press Win + X and open **Device Manager**. Expand **Mice and other pointing devices** and **Universal Serial Bus controllers** to check for error icons (yellow exclamation marks).

- **Driver Update/Reinstall:** Right-click the mouse/HID device and select **Update Driver** or **Uninstall Device**, then restart the system to allow Windows to auto-reinstall default drivers.
- **Bluetooth Settings (for Bluetooth Mice):** If using Bluetooth, go to *Settings > Bluetooth & devices*, remove the mouse from paired devices, and re-pair it.

Task 4:

Gaming PC Overheating & Auto-Shutdown Diagnosis

Objective: Identify 3 primary hardware causes for CPU/GPU overheating causing system thermal shutdown during intense gaming workloads, and describe verification procedures for each.

3 Hardware Causes and Verification Procedures:

1. Dust and Debris Accumulation in Heatsinks and Case Fans:

- **Hardware Cause:** Heavy dust accumulation on the CPU cooler fins, GPU radiator, and intake filters blocks essential airflow, trapping heat inside the chassis.
- **How to Check:** Open the side panel and perform a visual inspection of the CPU heatsink fins and

case fan blades. Use a flashlight to check if dust layer completely blocks the air passages.

2. Dried-Out or Degrading Thermal Paste (Thermal Interface Material):

- **Hardware Cause:** Over time, thermal paste between the CPU Integrated Heat Spreader (IHS) and the heatsink base dries up and loses thermal conductivity, causing rapid CPU temperature spikes ($>95^{\circ}\text{C}$) under load.
- **How to Check:** Unmount the CPU cooler/heatsink and inspect the thermal compound layer. Dried, flaky, or unevenly spread paste indicates an immediate need for cleaning with isopropyl alcohol and reapplication.

3. Cooling Fan Failure or Blocked AIO Liquid Pump:

- **Hardware Cause:** The CPU fan motor has failed/stuck, or in liquid-cooled systems, the AIO pump has stopped circulating coolant.
- **How to Check:** Power ON the system and visually observe if the fans are spinning. Boot

into BIOS or open monitoring software (HWMonitor/SpeedFan) to verify if the **CPU Fan RPM** displays 0 RPM or if the AIO pump tubes feel cold on one side and extremely hot on the other.